Scalability & Future Plan

Date	04 July 2026
Team ID	
Project Name	HDI Predictor Using Machine Learning
Maximum Marks	1 Mark

Scalability & Future Plan:
This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The application runs only on a local Flask server.	Limits accessibility for remote users.	High
2	The model is trained on a static HDI dataset and does not automatically update with new data.	Predictions may not reflect the latest development statistics.	Medium
3	The application supports single-country prediction only and lacks advanced analytical features.	Limits comparative analysis and broader decision support.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports local users on the Flask development server.	Deploy on a cloud platform (Render, Azure, or AWS) to support multiple concurrent users.
2	Data Storage	Uses a local HDI.csv dataset and HDI.pkl model file.	Integrate a database and periodically update the dataset from official sources.
3	Performance	Suitable for a small number of prediction requests.	Optimize model loading, implement caching, and improve backend efficiency.
4	Security	Basic input validation is implemented.	Add user authentication, HTTPS support, and secure deployment practices.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Deploy the application to a cloud platform for online access.	3–6 Months	Improve accessibility and availability.
Phase 3	Add data visualization dashboards, country comparison, and trend analysis.	6–9 Months	Provide richer analytical insights for users.

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 4	Integrate automatic dataset updates and enhance prediction accuracy using advanced machine learning models.	9–12 Months	Improve model performance and long-term usability.